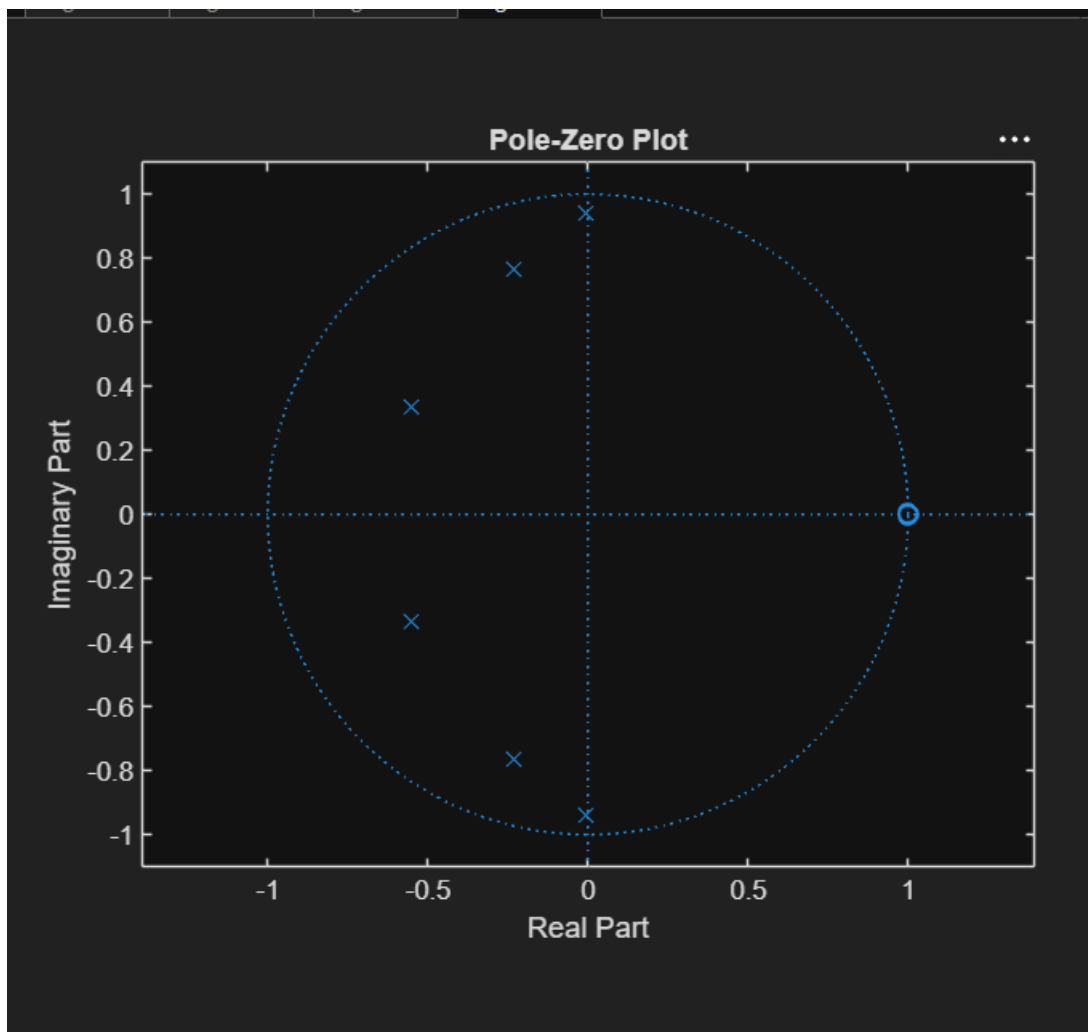
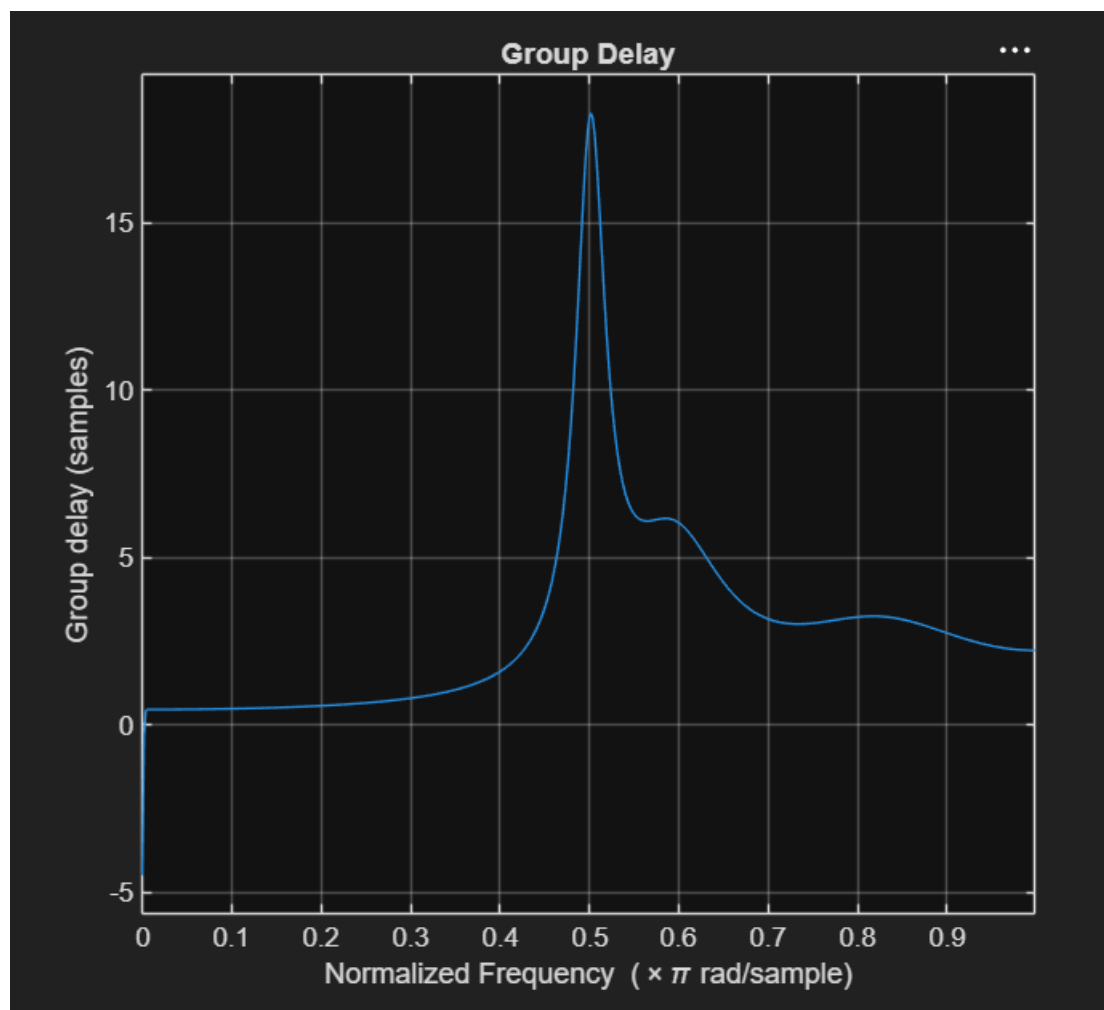
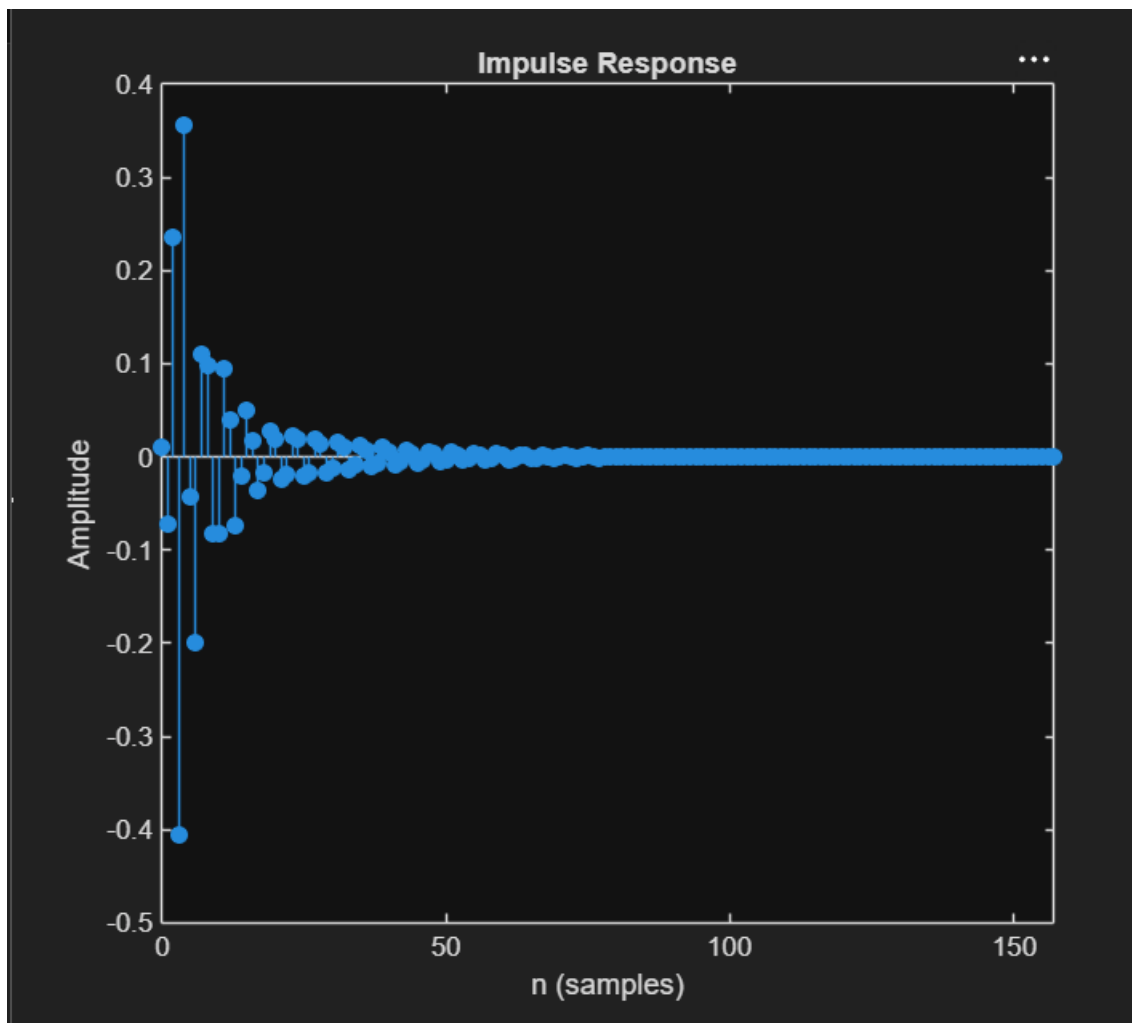
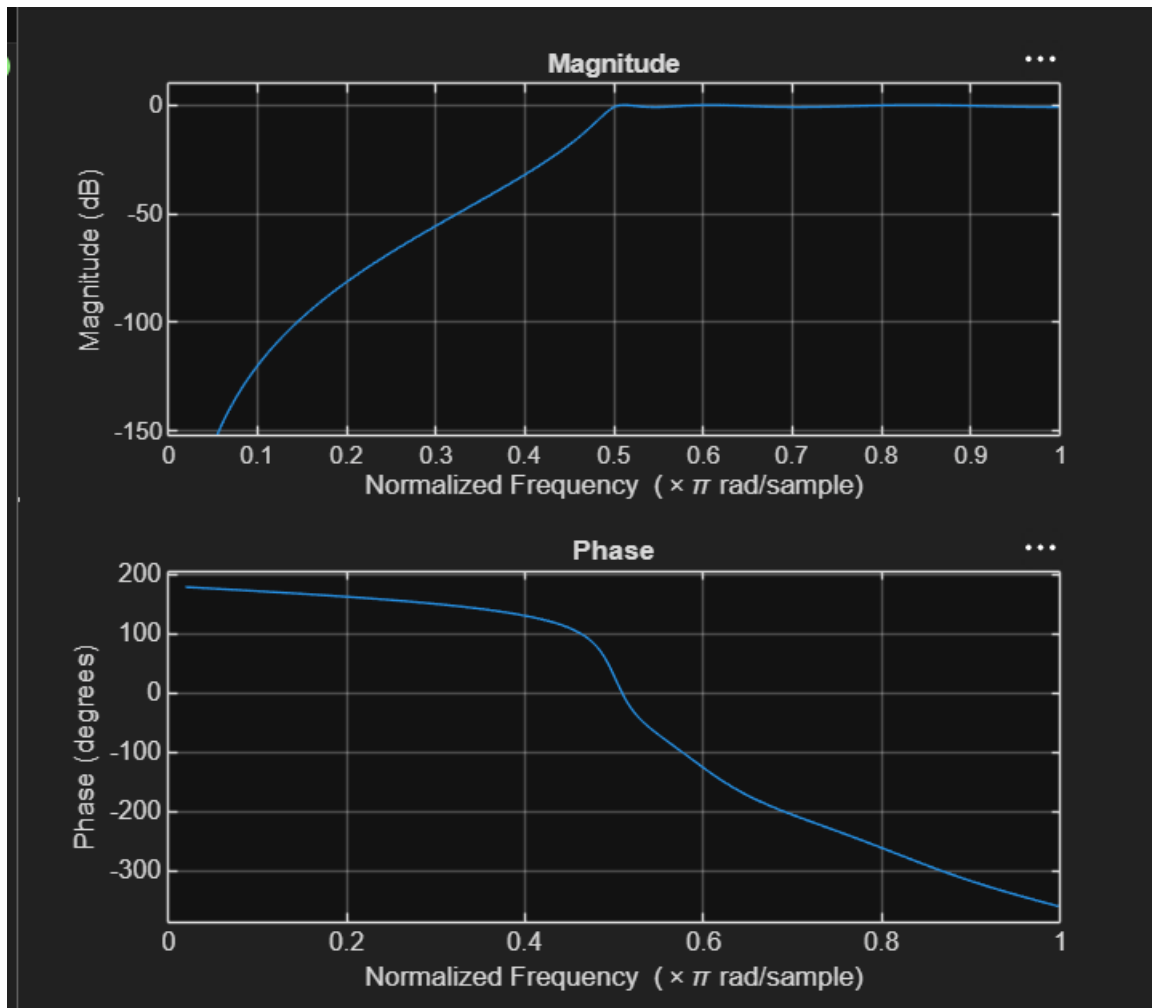










Program

```
clc;
```

```
clear;
```

```
close all;
```

```
% Specifications
```

```
Rp = 1; % Passband Ripple (dB)
```

```
Rs = 40; % Stopband Attenuation (dB)
```

```
Wp = 0.5; % Passband Edge
```

```
Ws = 0.35; % Stopband Edge
```

```
% Calculate filter order
```

```
[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);
```

```
% Design High-Pass Filter
```

```
[b,a] = cheby1(n,Rp,Wn,'high');
```

```
fprintf('Filter Order = %d\n',n);
```

```
fprintf('Cutoff Frequency = %.4f\n',Wn);
```

```
% Frequency Response
```

```
figure;
```

```
freqz(b,a);
```

```
% Impulse Response
```

```
figure;
```

```
impz(b,a);
```

```
% Group Delay
```

```
figure;
```

```
grpdelay(b,a);
```

```
% Pole-Zero Plot
```

```
figure;
```

```
zplane(b,a);
```